MOHAMED MESHRIF

Senior Flutter Engineer | Mobile Architecture | Android & iOS

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PROFESSIONAL SUMMARY

Senior Flutter Engineer with nearly 5 years building and shipping production apps for Android and iOS, with many live apps across marketplace, social, super-app, logistics, and news for clients in Egypt, Saudi Arabia, and Iraq. Takes features from architecture to the store: Clean Architecture + BLoC, platform integrations (subscriptions, background location), and Fastlane CI/CD. Author of 4 open-source pub.dev packages, including video_flux, a memory-safe preloader for short-video feeds. A team player and technical lead who guides teammates on architecture and code quality and works closely with designers, backend engineers, and product owners to ship on time.

TECHNICAL SKILLS

Languages & Frameworks: Dart, Flutter (Android, iOS, Web), Kotlin & Swift platform channels

Architecture: Clean Architecture, Feature-first, MVVM, SOLID, Repository & Use-case layers, Dependency Injection (get_it, injectable)

State Management: BLoC / Cubit, Riverpod, Provider, GetX

Networking & Data: REST APIs, Dio, Retrofit, Freezed, json_serializable, Hive, Sqflite, SharedPreferences

Backend & Cloud: Firebase (Auth, Firestore, FCM, Analytics, Crashlytics), Supabase

Platform Integrations: In-app purchases & subscriptions (StoreKit, Google Play Billing), background location & foreground services, Google Maps / flutter_map, payment gateways, deep links

Navigation & UI: GoRouter, responsive & adaptive layouts, RTL / localization, custom animations, Figma-to-Flutter

Testing: Unit, widget & integration tests, bloc_test, mocktail / mockito, golden tests, test coverage in CI

Performance & Quality: Flutter DevTools profiling, memory-leak & jank analysis, video preloading & controller lifecycle, code reviews, null-safety migration

DevOps & Tooling: Fastlane, GitHub Actions, flavors / multi-environment builds, Mason bricks, App Store & Play Store release

Leadership & Teamwork: Technical leadership, mentoring, code reviews, task estimation, cross-functional collaboration (design, backend, product)

AI-Assisted Engineering: Claude Code, Cursor, Gemini, custom agent rules & skills for team-wide code standards

PROFESSIONAL EXPERIENCE

Senior Flutter Developer | Aug 2025 – Present

Spark Systems – Programming Solutions | *El Sheikh Zayed, Egypt*

- **Architecture ownership:** Design the Flutter architecture for new product lines: feature-first Clean Architecture, BLoC/Cubit, use-case and repository layers, and DI with get_it/injectable, making every module independently testable.
- **Testing:** Cover business logic and UI with unit, bloc, and widget tests (bloc_test, mocktail) plus integration tests for critical flows, run automatically in the CI pipeline to catch regressions before release.
- **Team velocity:** Created Mason brick templates and project scaffolding that generate full feature modules in seconds, standardizing structure across the codebase and cutting setup time for new features.
- **Engineering standards:** Introduced AI-assisted development workflows (Claude Code rules & skills) that encode the team's architecture and coding conventions, improving review consistency.

Flutter Developer | Mar 2022 – Aug 2025

IT PLUS – Programming Solutions | *El Sheikh Zayed, Egypt*

- **Delivery:** Built and shipped many cross-platform apps to Google Play and the App Store across e-commerce, social, super-app, news, and logistics, collectively serving thousands of active users.
- **Migration lead:** Led the full native (Android/iOS) to Flutter rewrite of the flagship social-marketplace app (Itop), unifying two codebases into one while preserving full feature parity.
- **CI/CD:** Built and owned Fastlane pipelines for iOS and Android (automated build, code signing, versioning, and store submission), turning manual releases into a one-command process.
- **Real-time & location:** Engineered background GPS tracking and real-time driver–merchant sync for a logistics platform, stable across foreground/background transitions and optimized for battery.
- **Performance:** Profiled and fixed memory leaks, janky renders, and redundant network calls in inherited apps using Flutter DevTools, restoring stability and smooth UX.

- **Team standards:** Championed Clean Architecture, feature-first structure, DI, and BLoC as company-wide project standards; drove a large-scale null-safety migration.
- **Firebase:** Integrated Auth (incl. social login), Firestore, FCM push notifications, and Analytics across multiple production apps.

SELECTED PROJECTS

Itop | Social + Marketplace | Google Play · App Store
Stack: Flutter, BLoC, Firebase, Clean Architecture

- Led the native-to-Flutter migration; delivered voice recording, video playback, real-time feeds, and an in-app services marketplace.

Sela Super App | Saudi multi-vertical super app | Google Play · App Store
Stack: Flutter, Fastlane, modular architecture

- Added social, procurement, and logistics modules to a live super app with zero disruption to existing users; owned its Fastlane release pipeline.

Sela FHE (Falcon Expo) | Event app, Saudi Arabia | Google Play · App Store
Stack: Flutter, Clean Architecture, BLoC

- Built from scratch as sole Flutter engineer; worked directly with stakeholders on culturally specific design and backend API contracts.

Insta Elsouq | Marketplace | Google Play · App Store
Stack: Flutter, Google Maps, Firebase

- Dual-role (customer / vendor) platform with map-based service discovery, order tracking, social login, and push notifications.

Sareea Driver | Logistics driver app | App Store
Stack: Flutter, background location, foreground service

- Persistent background GPS tracking with live delivery updates, reliable across app-state transitions and battery-optimized for long shifts.

Observer Iraq | News | Google Play · App Store

- Category-based news with client-side caching for fast loads and offline reading.

Cinderella | Matchmaking | Google Play

- Took over an unstable app and fixed rendering bottlenecks, memory leaks, and network inefficiencies.

OPEN-SOURCE PACKAGES (PUB.DEV)

- **video_flux** (pub.dev/packages/video_flux): Player-agnostic sliding-window video preloader for TikTok/Reels-style feeds. Keeps only nearby controllers alive, adapts the window to device tier and memory pressure, debounces fast scrolls, and ships retry policies, lifecycle handling, and a diagnostics event stream. 160/160 pub points, all 6 platforms.
- **flavor_cli** (pub.dev/packages/flavor_cli): Dart CLI that automates Flutter flavor setup for Android and iOS (Gradle, Xcode schemes, configs), removing hours of manual multi-environment boilerplate.
- **resource_state_builder** (pub.dev/packages/resource_state_builder): Declarative widget for async resource states (loading / success / error / empty) with clean, type-safe UI handling.
- **app_sizer** (pub.dev/packages/app_sizer): Responsive sizing utility for consistent layouts across phones, tablets, and web without manual breakpoints.

EDUCATION

B.Sc. Electronics & Communications Engineering | 2015 – 2020
Zagazig University, Egypt

LANGUAGES

Arabic: Native | English: Professional working proficiency